

data_cleaning

September 17, 2026

```
[1]: # imports
import pandas as pd
import requests
import yfinance as yf
import numpy as np
from transformers import pipeline
from datetime import datetime, timedelta
import finnhub
from dotenv import load_dotenv
import os
from edgar import Company, set_identity
import regex as re
```

0.0.1 Gathering Data

```
[2]: headers = {"User-Agent": "Mozilla/5.0 (compatible; my-script/1.0)"}
url = 'https://en.wikipedia.org/wiki/List_of_S%26P_500_companies'
html = requests.get(url, headers=headers).text
table = pd.read_html(html)[0]
table['Symbol'] = table['Symbol'].str.replace('.', '-').tolist()
table.head()
```

/var/folders/5l/vsm1wvjn4qdg25qwjfcj83kr0000gn/T/ipykernel_99818/4004594965.py:4
: FutureWarning: Passing literal html to 'read_html' is deprecated and will be removed in a future version. To read from a literal string, wrap it in a 'StringIO' object.

```
table = pd.read_html(html)[0]
```

```
[2]:
```

	Symbol	Security	GICS Sector	\
0	MMM	3M	Industrials	
1	AOS	A. O. Smith	Industrials	
2	ABT	Abbott Laboratories	Health Care	
3	ABBV	AbbVie	Health Care	
4	ACN	Accenture	Information Technology	

	GICS Sub-Industry	Headquarters Location	Date added	\
0	Industrial Conglomerates	Saint Paul, Minnesota	1957-03-04	
1	Building Products	Milwaukee, Wisconsin	2017-07-26	

2	Health Care Equipment	North Chicago, Illinois	1957-03-04
3	Biotechnology	North Chicago, Illinois	2012-12-31
4	IT Consulting & Other Services	Dublin, Ireland	2011-07-06

	CIK	Founded
0	66740	1902
1	91142	1916
2	1800	1888
3	1551152	2013 (1888)
4	1467373	1989

Tickers Semiconductors: NVDA (NVIDIA), AMD (Advanced Micro Devices), INTC (Intel), ASML (ASML Holding), TSM (Taiwan Semiconductor Manufacturing), AVGO (Broadcom), MRVL (Marvell Technology), ARM (Arm Holdings), SMC (Super Micro Computer) **Hyperscalers:** MSFT (Microsoft), GOOGL (Google), AMZN (Amazon), META (Facebook) **AI Pure Plays:** PLTR (Palantir), AI (C3.ai), SOUN (Soundhound AI) **Infrastructure:** DELL (Dell Technologies), ANET (Arista Networks), HPE (Hewlett Packard Enterprise)

```
[3]: # list of all the tickers with a dictionary that sorts them by type of stock
tickers = pd.read_csv('../data/tickers.csv')
tickers = tickers.drop(columns=['Unnamed: 0'])

ai_tickers = {
    'semis': ['NVDA', 'AMD', 'INTC', 'ASML', 'TSM', 'AVGO', 'MRVL', 'ARM', 'SMCI'], #_
    ↪semiconductors
    'hypers': ['MSFT', 'GOOGL', 'AMZN', 'META'], # hyperscalar
    'pure': ['PLTR', 'AI', 'SOUN'], # AI pure plays
    'infra': ['DELL', 'ANET', 'HPE'] # infrastructure
}
all_tickers = [t for group in ai_tickers.values() for t in group]
```

```
[4]: # Data from 2018 - 2026 (pre-COVID to present)
prices = yf.download(
    tickers=all_tickers + ['SOXX', 'QQQ'],
    start='2018-01-01',
    end='2026-01-01',
    auto_adjust=True,
    progress=False
)
prices.to_csv('../data/prices_raw.csv')
```

```
[5]: prices.head()
```

```
[5]: Price      Close
Ticker      AI      AMD      AMZN      ANET ARM      ASML      AVGO      \
Date
2018-01-02  NaN  10.98  59.450500  14.439375 NaN  163.474487  21.053339
```

2018-01-03	NaN	11.55	60.209999	14.725000	NaN	164.716141	21.283579
2018-01-04	NaN	12.12	60.479500	14.543125	NaN	166.252228	21.290680
2018-01-05	NaN	11.88	61.457001	14.798125	NaN	168.202194	21.416836
2018-01-08	NaN	12.28	62.343498	15.691250	NaN	169.085190	21.468090

Price				...	Volume		
Ticker	DELL	GOOGL	HPE	...	META	MRVL	MSFT
Date				...			
2018-01-02	21.014696	53.188862	11.300007	...	18151900	5619900	22483800
2018-01-03	21.236488	54.096317	11.369617	...	16886600	9014000	26061400
2018-01-04	21.613806	54.306446	11.648057	...	13880900	9221200	21912000
2018-01-05	21.771868	55.026569	11.640323	...	13574500	5855400	23407100
2018-01-08	21.766768	55.220844	11.462433	...	17994700	5529500	22113000

Price							
Ticker	NVDA	PLTR	QQQ	SMCI	SOUN	SOXX	TSM
Date							
2018-01-02	355616000	NaN	32573300	4344000	NaN	2860200	4984000
2018-01-03	914704000	NaN	29383600	4094000	NaN	1326900	6963200
2018-01-04	583268000	NaN	24776100	3631000	NaN	1335000	4876600
2018-01-05	580124000	NaN	26992300	2178000	NaN	1170600	5330800
2018-01-08	881216000	NaN	23159100	3484000	NaN	1622700	3538200

[5 rows x 105 columns]

0.0.2 Cleaning Data

```
[6]: close = prices['Close']
      volume = prices['Volume']
      soxx = close['SOXX']
      qqq = close['QQQ']
```

```
[7]: # Remove tickers with more than 5% NaN values
      clean_prices = close.loc[:, close.isna().mean() <= 0.05]

      # Forward fill short gaps of 1-2 days
      clean_prices = clean_prices.ffill(limit=2)
      clean_prices = clean_prices.dropna()
      clean_prices.to_csv('../data/clean_prices.csv')
```

```
[8]: # Convert prices to log
      log_ret = np.log(clean_prices / clean_prices.shift(1)).dropna()
```

0.0.3 Feature Engineering

```
[9]: # SOXX: holds 30 largest semiconductor companies
# QQQ: holds 100 largest non-financial companies on the NASDAQ, heavily
    ↪ weighted towards big tech
def make_features(close: pd.Series,
                  volume: pd.Series,
                  soxx: pd.Series,
                  qqq: pd.Series) -> pd.DataFrame:
    df = pd.DataFrame(index=close.index)

    # Lag returns: percent change over x days
    for lag in [1, 3, 5, 10, 20]:
        df[f'ret_{lag}d'] = close.pct_change(lag)

    # Rolling volatility
    df['vol_5d'] = close.pct_change().rolling(5).std()
    df['vol_20d'] = close.pct_change().rolling(20).std()

    # RSI (14-day): Relative Strength Index; measures magnitude and recent
    ↪ price changes in stocks
    delta = close.pct_change()
    gain = delta.clip(lower=0).rolling(14).mean()
    loss = (-delta.clip(upper=0)).rolling(14).mean()
    df['rsi_14'] = 100 - (100 / (1 + gain / loss))

    # How does the stock compare to SOXX and QQQ?
    df['rel_soxx'] = close.pct_change(20) - soxx.pct_change(20)
    df['rel_qqq'] = close.pct_change(20) - qqq.pct_change(20)

    # Volume spike: number of shares traded on a given day
    df['vol_spike'] = volume / volume.rolling(20).mean()

    # 52-week high proximity: the highest price traded at over the last 252
    ↪ trading days (1 year)
    df['pct_from_52wk_high'] = close / close.rolling(252).max() - 1

    return df.dropna()
```

```
[10]: all_feature_dfs = []
for ticker in all_tickers:
    if ticker not in close.columns:
        continue
    ticker_feats = make_features(
        close = close[ticker],
        volume = volume[ticker],
        soxx = soxx,
```

```

        qqq      = qqq
    )
    ticker_feats['Ticker'] = ticker
    all_feature_dfs.append(ticker_feats)

```

```

[11]: features_df = pd.concat(all_feature_dfs).reset_index()
      features_df = features_df.set_index(['Date', 'Ticker'])

```

```

[12]: # Forward 5-day cumulative return
      features_df['fwd_5d'] = (
          features_df.groupby('Ticker')['ret_1d']
          .transform(lambda x: x.shift(-1).rolling(5).sum())
      )

      # Cross-sectional: top half = 1, bottom half = 0 (guaranteed 50/50 per date)
      # Which stocks will outperform their peers?
      features_df['Target'] = (
          features_df.groupby('Date')['fwd_5d']
          .transform(lambda x: (x > x.median()).astype(int))
      )

```

```

/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-
packages/numpy/lib/_nanfunctions_impl.py:1241: RuntimeWarning: Mean of empty
slice
    return np.nanmean(a, axis, out=out, keepdims=keepdims)
/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-
packages/numpy/lib/_nanfunctions_impl.py:1241: RuntimeWarning: Mean of empty
slice
    return np.nanmean(a, axis, out=out, keepdims=keepdims)
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packages/numpy/lib/_nanfunctions_impl.py:1241: RuntimeWarning: Mean of empty
slice
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/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-
packages/numpy/lib/_nanfunctions_impl.py:1241: RuntimeWarning: Mean of empty
slice
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/Library/Frameworks/Python.framework/Versions/3.12/lib/python3.12/site-
packages/numpy/lib/_nanfunctions_impl.py:1241: RuntimeWarning: Mean of empty
slice
    return np.nanmean(a, axis, out=out, keepdims=keepdims)

```

```

[13]: features_df.head()

```

```

[13]:
      ret_1d  ret_3d  ret_5d  ret_10d  ret_20d  vol_5d  \
Date      Ticker
2019-01-02 NVDA    0.020375  0.038500  0.071923 -0.051261 -0.166494  0.023554
2019-01-03 NVDA   -0.060417 -0.042350 -0.038392 -0.128964 -0.247295  0.033031

```

2019-01-04	NVDA	0.064068	0.020150	0.038271	-0.016750	-0.133155	0.045219
2019-01-07	NVDA	0.052941	0.052709	0.072952	0.061436	-0.094068	0.049557
2019-01-08	NVDA	-0.024896	0.092507	0.047416	0.079185	-0.052706	0.052560

Date	Ticker	vol_20d	rsi_14	rel_soxx	rel_qqq	vol_spike	\
2019-01-02	NVDA	0.034328	37.207888	-0.107088	-0.083557	0.785101	
2019-01-03	NVDA	0.034169	30.211087	-0.110360	-0.119161	1.104590	
2019-01-04	NVDA	0.035031	40.655574	-0.080480	-0.078587	0.933377	
2019-01-07	NVDA	0.037384	48.847622	-0.063841	-0.044049	1.128762	
2019-01-08	NVDA	0.034712	48.297116	-0.055655	-0.044009	1.240790	

Date	Ticker	pct_from_52wk_high	fwd_5d	Target
2019-01-02	NVDA	-0.528766	NaN	0
2019-01-03	NVDA	-0.557237	NaN	0
2019-01-04	NVDA	-0.528870	NaN	0
2019-01-07	NVDA	-0.503928	NaN	0
2019-01-08	NVDA	-0.516278	0.051362	1

```
[14]: features_df.to_csv('../data/features_df.csv')
```

0.0.4 500 Most Recent News Sentiment with Finnhub

```
[15]: load_dotenv()
```

```
[15]: True
```

```
[16]: client = finnhub.Client(api_key=os.getenv("FINNHUB_API_KEY"))
```

```
[17]: # date, headline, and summary of most recent 500 news per ticker
def get_ai_news(ticker, start_date, end_date):
    articles = []
    news = client.company_news(ticker, _from=start_date, to=end_date)
    for item in news[:500]:
        articles.append({
            'date': datetime.fromtimestamp(item['datetime']).date(),
            'headline': item['headline'],
            'summary': item.get('summary', '')
        })
    return pd.DataFrame(articles)
```

```
[18]: finbert = pipeline('sentiment-analysis', model='ProsusAI/finbert',
    ↪truncation=True, max_length=512)
```

Device set to use mps:0

```
[19]: def clean_filing_text(text):
    text = text.lower()
    text = re.sub(r'<[>]+>', ' ', text)           # remove HTML tags
    text = re.sub(r'&[a-z]+;', ' ', text)           # remove HTML entities
    text = re.sub(r'\s+', ' ', text)                 # normalize whitespace
    text = re.sub(r'\d{1,2}/\d{1,2}/\d{2,4}', '', text) # remove dates
    text = re.sub(r'\$[\d,\.]+', '', text)           # remove dollar amounts
    text = re.sub(r"'", '', text)                   # drop apostrophes (It's ->
↳ Its), no gap left behind
    text = re.sub(r'[\w\s\.]', ' ', text)           # remove special characters
    # text = re.sub(r'\s+', ' ', text)               # normalize whitespace
    text = re.sub(r'\s{2,}', ' ', text)
    return text.strip()
```

```
[20]: def clean_headlines(text):
    text = re.sub(r'<[>]+>', ' ', text)           # remove HTML tags
    text = re.sub(r'\s+', ' ', text)                 # normalize whitespace
    text = re.sub(r'[\w\s\.]', ' ', text)           # remove special characters
    return text.strip()
```

```
[21]: def score_headlines(df):
    texts = (df['headline'] + ' . ' + df['summary']).tolist()
    results = finbert(texts)
    df['sentiment'] = [r['label'].lower() for r in results]
    df['score'] = [r['score'] if r['label'] == 'positive' else -r['score'] if
↳ r['label'] == 'negative' else 0 for r in results]
    daily = df.groupby('date').agg(
        news_sentiment = ('score', 'mean'),
        news_count = ('score', 'count'),
        pct_positive = ('sentiment', lambda x: (x=='positive').mean())
    )
    return daily
```

```
[22]: def get_sentiment_for_ticker(ticker, df_ticker):
    # find the beginning and end dates for each ticker
    start_date = df_ticker.index.get_level_values('Date').min().
↳ strftime("%Y-%m-%d")
    end_date = df_ticker.index.get_level_values('Date').max().
↳ strftime("%Y-%m-%d")

    news_df = get_ai_news(ticker, start_date, end_date)
    if news_df.empty:
        return pd.DataFrame()

    # clean text
    news_df['headline'] = news_df['headline'].apply(clean_headlines)
    # news_df['summary'] = news_df['summary'].apply(clean_filing_text)
```

```

# provides sentiment rating based on 500 most recent news for each ticker
daily_sentiment = score_headlines(news_df)
daily_sentiment['Ticker'] = ticker
return daily_sentiment

```

```

[23]: # Loop over all tickers
all_sentiment = []
for ticker in features_df.index.get_level_values('Ticker').unique():
    df_ticker = features_df.xs(ticker, level='Ticker')
    sentiment = get_sentiment_for_ticker(ticker, df_ticker)
    if not sentiment.empty:
        all_sentiment.append(sentiment)

sentiment_df = pd.concat(all_sentiment)
sentiment_df.index = pd.to_datetime(sentiment_df.index)
sentiment_df.to_csv('../data/sentiment_df.csv')

```

```

[24]: # merge sentiment df back to features df
df = features_df.reset_index()
df['Date'] = pd.to_datetime(df['Date'])
df = df.merge(
    sentiment_df.reset_index().rename(columns={'date': 'Date'}),
    on=['Date', 'Ticker'],
    how='left'
)
df = df.set_index(['Date', 'Ticker'])

# Fill missing sentiment (days with no news)
df[['news_sentiment', 'new_count', 'pct_positive']] = df[['news_sentiment',
↪ 'new_count', 'pct_positive']].fillna(0)

```

0.0.5 SEC 8K Filings (Earnings Releases)

```

[25]: # Domain keyword dictionaries
NVIDIA_KEYWORDS = [
    'h100', 'h200', 'h20',
    'blackwell', 'b100', 'b200', 'gb200', 'gb300',
    'rubin', 'vera rubin', 'nvl72',
    'dgx', 'hgx',
    'cuda', 'tensorrt', 'triton', 'nim microservices',
    'hopper architecture', 'blackwell architecture',
]

AMD_KEYWORDS = [
    'instinct mi300', 'instinct mi300x', 'instinct mi325',

```

```

'instinct mi350', 'instinct mi400',
'instinct mi250', 'instinct mi200',
'radeon pro', 'radeon instinct',
'rocm', 'hip framework', 'amd cdna',
'amd rdna', 'amd infinity fabric',
'amd gpu demand', 'amd accelerator',
'amd data center gpu', 'amd ai accelerator',
'amd compute demand', 'amd inference',
'amd training workload', 'amd cluster',
]

INTEL_KEYWORDS = [
'gaudi2', 'gaudi 2', 'gaudi3', 'gaudi 3',
'habana labs', 'habana gaudi',
'ponte vecchio', 'max series gpu',
'intel gpu max', 'flex series gpu',
'openvino', 'intel oneapi', 'intel neural compressor',
'intel xeon ai', 'intel core ultra ai',
'meteor lake ai', 'lunar lake ai', 'arrow lake ai', 'intel npu',
'intel data center gpu', 'intel ai accelerator',
'intel compute demand', 'intel ai inference',
'intel training workload', 'gaudi deployment', 'gaudi adoption', 'gaudi_
demand',
]

GENERAL_GPU_KEYWORDS = [
'gpu demand', 'gpu supply', 'gpu allocation',
'data center revenue', 'data center demand',
'inference demand', 'inference workload',
'training compute', 'compute demand',
'cloud ai', 'ai infrastructure', 'ai accelerator',
'tensor processing', 'custom ai chip',
]

CAPEX_UP_KEYWORDS = [
'increasing capex', 'raised capex', 'accelerating capex',
'accelerating investment', 'capital expenditure increase',
'expanding capacity', 'record investment',
'investing heavily', 'multiyear investment',
'committed to investing', 'significant investment in ai',
'capital investment', 'infrastructure investment', 'ai investment'
]

CAPEX_DOWN_KEYWORDS = [
'reducing capex', 'cutting investment', 'slowing spend',
'deferring investment', 'pausing investment',

```

```

    'capital efficiency', 'disciplined spending',
    'revisiting capex', 'moderating investment'
]

COMPETITIVE_RISK_KEYWORDS = [
    'amd instinct', 'gaudi', 'trainium', 'inferentia', 'maia',
    'google tpu', 'axion',
    'custom silicon', 'in-house chip', 'in-house ai chip',
    'merchant silicon', 'proprietary chip',
    'competitive alternative', 'alternative to nvidia',
    'amd alternative', 'nvidia alternative',
    'vendor diversification', 'multi vendor gpu',
]

```

```

[26]: def extract_domain_signals(transcript_text):
    text_lower = transcript_text.lower()
    signals = {}

    # Raw mention counts (normalize by transcript length)
    word_count = len(text_lower.split())
    signals['nvidia_mentions'] = sum(text_lower.count(k) for k in
↳NVIDIA_KEYWORDS) / word_count * 1000
    signals['amd_mentions'] = sum(text_lower.count(k) for k in AMD_KEYWORDS) /
↳word_count * 1000
    signals['intel_mentions'] = sum(text_lower.count(k) for k in
↳INTEL_KEYWORDS) / word_count * 1000
    signals['gpu_mentions'] = sum(text_lower.count(k) for k in
↳GENERAL_GPU_KEYWORDS) / word_count * 1000
    signals['capex_up_score'] = sum(text_lower.count(k) for k in
↳CAPEX_UP_KEYWORDS) / word_count * 1000
    signals['capex_down_score'] = sum(text_lower.count(k) for k in
↳CAPEX_DOWN_KEYWORDS) / word_count * 1000
    competitor_keys = [k for k in COMPETITIVE_RISK_KEYWORDS if k != ticker.
↳lower()]
    signals['competitor_mentions'] = sum(text_lower.count(k) for k in
↳competitor_keys) / word_count * 1000
    signals['capex_net'] = signals['capex_up_score'] -
↳signals['capex_down_score']

    # Paragraph-level: find sentences mentioning AI demand
    ai_sentences = [s for s in transcript_text.split('.')
        if any(k in s.lower() for k in ['artificial_
↳intelligence', 'ai demand', 'ai revenue', 'ai growth', 'ai contribution'])]
    signals['ai_sentence_ratio'] = len(ai_sentences) / max(len(transcript_text.
↳split('.')), 1)

```

```
return signals
```

```
[27]: load_dotenv()
```

```
[27]: True
```

```
[28]: set_identity(os.getenv('EMAIL'))
```

```
[29]: # get earnings transcript
def get_earnings_transcripts(ticker):
    company = Company(ticker)
    filings = company.get_filings(form="8-K")

    results = []
    for filing in filings:
        try:
            # only earnings releases
            if '2.02' not in str(filing.items):
                continue

            for attachment in filing.attachments:
                if 'EX-99' not in attachment.document_type:
                    continue

                text = attachment.text()

                if not text or len(text) < 200:
                    continue

                text = clean_filing_text(text) # reuse cleaning text regex
                signals = extract_domain_signals(text)

                results.append({
                    'Ticker': ticker,
                    'Date': pd.to_datetime(filing.filing_date),
                    'Text': text,
                    **signals
                })
                break # one exhibit per filing

        except Exception as e:
            print(f"Skipped {ticker} {filing.filing_date}: {e}")
            continue

    print(f"{ticker}: {len(results)} qualifying filings")
    return results
```

```

[30]: all_transcript_signals = []

for ticker in all_tickers:
    transcripts = get_earnings_transcripts(ticker)
    for t in transcripts:
        signals = extract_domain_signals(t['Text'])
        signals['Ticker'] = t['Ticker']
        signals['Date'] = t['Date']
        all_transcript_signals.append(signals)

transcript_signals_df = (pd.DataFrame(all_transcript_signals).
    ↳set_index(['Date', 'Ticker']).sort_index())

transcript_signals_df.to_csv('../data/transcript_signals_df.csv')
transcript_signals_df.head()

```


NVDA: 94 qualifying filings

AMD: 95 qualifying filings

INTC: 109 qualifying filings
ASML: 0 qualifying filings
TSM: 0 qualifying filings

AVGO: 34 qualifying filings

MRVL: 24 qualifying filings
ARM: 0 qualifying filings

SMCI: 101 qualifying filings

MSFT: 90 qualifying filings

GOOGL: 45 qualifying filings

AMZN: 88 qualifying filings

META: 57 qualifying filings

PLTR: 25 qualifying filings

AI: 26 qualifying filings

SOUN: 19 qualifying filings

DELL: 42 qualifying filings

ANET: 49 qualifying filings

HPE: 54 qualifying filings

```
[30]:
```

Date	Ticker	nvidia_mentions	amd_mentions	intel_mentions	\
2004-10-07	AMD	0.0	0.0	0.0	
2004-10-12	INTC	0.0	0.0	0.0	
2004-10-21	AMZN	0.0	0.0	0.0	
	MSFT	0.0	0.0	0.0	
2004-10-27	NVDA	0.0	0.0	0.0	

Date	Ticker	gpu_mentions	capex_up_score	capex_down_score	\
2004-10-07	AMD	0.0	0.0	0.0	
2004-10-12	INTC	0.0	0.0	0.0	
2004-10-21	AMZN	0.0	0.0	0.0	
	MSFT	0.0	0.0	0.0	
2004-10-27	NVDA	0.0	0.0	0.0	

Date	Ticker	competitor_mentions	capex_net	ai_sentence_ratio
2004-10-07	AMD	0.0	0.0	0.0
2004-10-12	INTC	0.0	0.0	0.0
2004-10-21	AMZN	0.0	0.0	0.0
	MSFT	0.0	0.0	0.0
2004-10-27	NVDA	0.0	0.0	0.0

```
[31]: nvda_tickers = get_earnings_transcripts('NVDA')
      nvda_tickers[0]
```


NVDA: 94 qualifying filings

```
[31]: {'Ticker': 'NVDA',  
      'Date': Timestamp('2026-08-26 00:00:00'),  
      'Text': 'nvidia announces financial results for second quarter fiscal 2027  
revenue of billion up 106 from a year ago data center revenue of billion up 117  
from a year ago santa clara calif. aug. 26 2026 nvidia nasdaq nvda today  
reported revenue for the second quarter ended july 26 2026 of billion up 18 from  
the previous quarter and up 106 from a year ago. for the quarter gaap and non  
gaap gross margins were both 75.0 . gaap and non gaap earnings per diluted share  
were and respectively. ai has reached its inflection point. its doing useful  
work. its tokens are productive and profitable. now compute is revenue said  
jensen huang founder and ceo of nvidia. and demand is accelerating. this time
```

last year one lab alone was driving the buildout today we have a golden age of new ai labs and startups multiple frontier labs scaling in parallel a thriving open model ecosystem and physical ai coming online with strong momentum across the u.s. and around the world. the ai infrastructure buildout is at full steam. vera rubin now in full production was built to power exactly this moment. during the second quarter of fiscal 2027 nvidia returned approximately billion to shareholders in the form of shares repurchased and cash dividends. as of the end of the second quarter the company had approximately billion remaining under its share repurchase authorization. nvidia will pay its next quarterly cash dividend of per share on october 1 2026 to all shareholders of record on september 10 2026. q2 fiscal 2027 summary gaap in mill exce q2 q1 fy27 q2 fy26 q q y y earn fy27 per shar reve 18 106 gross 75.0 74.9 72.4 0.1 pts 2.6 marg pts oper 10 55 expe oper 19 124 inco net 2 126 inco dilu earn 3 128 per share non g in milli except q2 fy27 q1 fy27 q2 fy26 q q y y earni per share reven 18 106 gross 75.0 75.0 72.5 2.5 pts margin opera 11 54 expen opera 19 124 income net 18 118 income dilut earni 19 120 per share outlook nvidias outlook for the third quarter of fiscal 2027 is as follows revenue is expected to be billion plus or minus 2 . nvidia is not assuming any data center compute revenue from china in its outlook. gaap and non gaap gross margins are expected to be 74.0 plus or minus 50 basis points. gaap and non gaap operating expenses are expected to be approximately billion and billion respectively. for the full year fiscal 2027 nvidia expects gaap and non gaap tax rates to be between 16.0 and 18.0 excluding any discrete items and material changes to nvidias tax environment. highlights data center second quarter revenue was billion up 18 from the previous quarter and up 117 from a year ago. announced the nvidia vera rubin platform is ramping into full production with racks running at partners including coreweave google cloud microsoft azure oracle cloud infrastructure and nebius. revealed that nvidia spectrum 6 switch systems supporting both pluggable and co packaged optics as part of the nvidia vera rubin platform are arriving across the worlds gigascale ai factories. unveiled nvidia vera the first cpu built for ai agents with broad adoption planned across the worlds leading technology providers. announced that nvidia groq 3 lpx the interactive ai inference accelerator is now in full production. introduced new security innovations for nvidia vera bluefield 4 stx delivering agentic ai storage processing with in silicon security for ai factories. launched the nvidia dsx platform providing infrastructure builders a complete playbook to design build and operate ai factories at scale. announced strategic partnerships to establish independent compute financing platforms with apollo blackrock blackstone brookfield goldman sachs and kkr to mobilize over billion of third party capital for the buildout of ai infrastructure over time subject to definitive agreements. announced that spacexai will deploy nvidia vera cpus to accelerate its next generation of agentic ai applications. secured land power and shell capacity through a partnership with sb energy at the ports pike technology campus in ohio to host nvidia compute. revealed that nvidia blackwell led across every category in the mlperf training 6.0 benchmarks and in agentperf the industrys first agentic ai infrastructure benchmark. announced new software open source models and partnerships with the worlds leading software platform providers to build

autonomous ai agents for industries and enterprises expanded nvidia agent toolkit with nvidia physicsnemo and new and updated nvidia cuda x libraries. launched nvidia bionemo agent toolkit which provides domain specific tools and skills for the agentic life sciences era. formed the open secure ai alliance with industry leaders to advance ai safety and security. revealed that nvidia gpus with confidential computing are now used for confidential inference in apples private cloud compute. expanded koreas ai factory ecosystem through strategic partnerships with sk telecom naver and brookfield to build sovereign ai infrastructure at gigawatt scale on the nvidia dsx platform as part of a broader nvidia powered national ai push across koreas industries and research institutions. announced a multiyear technology partnership with sk hynix to advance next generation memory for the global ai factory buildout. partnered with the japan government and industrial leaders to launch the worlds first national ai infrastructure revealed that japans leading enterprises startups and research institutions are building industry specialized ai models with nvidia nemotron open models. announced that the nvidia vera rubin platform delivers world class supercomputers for science announced a record 35 new nvidia ai hpc supercomputers in development across europe. edge computing second quarter edge computing revenue was billion up 13 from the previous quarter and up 27 from a year ago. partnered with microsoft to reinvent the windows pc with nvidia rtx spark a 1 petaflop superchip with the full cuda and nvidia rtx ecosystem. announced nvidia dgx station for windows the worlds most powerful desktide ai supercomputer for developing and running agents on windows. launched a local ai initiative with optimizations for top open models deepseek v4 flash diffusion gemma nemotron 3.5 lightning and qwen 3.8 and agent harnesses hermes agent and openclaw across rtx and dgx platforms. expanded the nvidia drive hyperion robotaxi ready platform ecosystem including strategic collaborations with foxconn vinfast uber and humain. introduced nvidia alpamayo 2 super the frontier open reasoning model for safe robotaxi and autonomous vehicle development for commercial use. launched nvidia cosmos 3 the worlds first fully open frontier omnimodel for physical ai. announced the nvidia isaac gr00t reference humanoid robot the first open humanoid robot reference design built on nvidia jetson thor and the nvidia isaac gr00t open development platform. announced nvidia halos for robotics the industrys first full stack safety system for physical ai unifying ai compute and safety in a single comprehensive platform. released a major collection of open source agent tools and skills for physical ai enabling developers to turn complex robotics autonomous vehicle and vision ai workflows into agent executable tasks. cfo commentary commentary on the quarter by colette kress nvidias executive vice president and chief financial officer is available at <https://investor.nvidia.com>. conference call and webcast information nvidia will conduct a conference call with analysts and investors to discuss its second quarter fiscal 2027 financial results and current financial prospects today at 2 p.m. pacific time 5 p.m. eastern time . a live webcast listen only mode of the conference call will be accessible at nvidias investor relations website <https://investor.nvidia.com>. the webcast will be recorded and available for replay until nvidias conference call to discuss its financial results for its third quarter of fiscal 2027. non gaap measures to supplement nvidias condensed consolidated

financial statements presented in accordance with gaap the company uses non gaap measures of certain components of financial performance. these non gaap measures include non gaap gross profit non gaap gross margin non gaap operating expenses non gaap operating income non gaap other income expense net non gaap net income non gaap net income or earnings per diluted share and free cash flow. for nvidias investors to be better able to compare its current results with those of previous periods the company has shown a reconciliation of gaap to non gaap financial measures. these reconciliations adjust the related gaap financial measures to exclude acquisition related and other costs other gains losses from equity securities net certain other income and expense and the associated tax impact of these items where applicable. beginning in the first quarter of fiscal 2027 nvidias non gaap financial measures no longer exclude stock based compensation expense. the historical non gaap financial information presented has been updated to include stock based compensation expense. free cash flow is calculated as gaap net cash provided by operating activities less both purchases related to property and equipment and intangible assets and principal payments on property and equipment and intangible assets. nvidia believes the presentation of its non gaap financial measures enhances the users overall understanding of the companys historical financial performance. the presentation of the companys non gaap financial measures is not meant to be considered in isolation or as a substitute for the companys financial results prepared in accordance with gaap and the companys non gaap measures may be different from non gaap measures used by other companies. about nvidia nvidia nasdaq nvda is the world leader in ai and accelerated computing. for further information contact toshiya hari mylene mangalindan investor relations corporate communications nvidia corporation nvidia corporation ir nvidia.com press nvidia.com certain statements in this press release including but not limited to statements as to the buildout of ai factories expectations with respect to growth performance and benefits of nvidias products services and technologies including blackwell and vera rubin and related trends and drivers expectations with respect to supply and demand for nvidias products services and technologies including blackwell and vera rubin and related matters including inventory production and distribution expectations with respect to nvidias third party arrangements including with its collaborators and partners expectations with respect to our investments expectations with respect to our financing arrangements expectations with respect to technology developments and related trends and drivers future nvidia cash dividends or other returns to stockholders nvidias financial and business outlook for the third quarter of fiscal 2027 and beyond projected market growth and trends expectations with respect to ai and related industries and other statements that are not historical facts are forward looking statements within the meaning of section 27a of the securities act of 1933 as amended and section 21e of the securities exchange act of 1934 as amended which are subject to the safe harbor created by those sections based on managements beliefs and assumptions and on information currently available to management and are subject to risks and uncertainties that could cause results to be materially different than expectations. important factors that could cause actual results to differ materially include global economic and political

conditions nvidias reliance on third parties to manufacture assemble package and test nvidias products the impact of technological development and competition development of new products and technologies or enhancements to nvidias existing products and technologies market acceptance of nvidias products or nvidias partners products design manufacturing or software defects changes in consumer preferences or demands changes in industry standards and interfaces unexpected loss of performance of nvidias products or technologies when integrated into systems nvidias ability to realize the potential benefits of business investments or acquisitions and changes in applicable laws and regulations as well as other factors detailed from time to time in the most recent reports nvidia files with the securities and exchange commission or sec including but not limited to its annual report on form 10 k and quarterly reports on form 10 q. copies of reports filed with the sec are posted on the companys website and are available from nvidia without charge. these forward looking statements are not guarantees of future performance and speak only as of the date hereof and except as required by law nvidia disclaims any obligation to update these forward looking statements to reflect future events or circumstances. 2026 nvidia corporation. all rights reserved. nvidia the nvidia logo bluefield cosmos nemotron nvidia isaac bionemo cuda x nvidia drive hyperion nvidia spectrum nvidia dsx nvidia rtx rtx spark dgx station and jetson thor are trademarks and or registered trademarks of nvidia corporation in the u.s. and or other countries. other company and product names may be trademarks of the respective companies with which they are associated. features pricing availability and specifications are subject to change without notice. nvidia corporation condensed consolidated statements of income in millions except per share data unaudited three six months months ended ended 2026 july 26 2025 july 27 2026 july 26 2025 july 27 revenue 96 221 46 743 177 837 90 805 cost of 24 079 12 890 44 538 30 284 revenue gross 72 142 33 853 133 299 60 521 profit operat expens research and 7 054 4 291 13 375 8 280 develop sales general 1 354 1 122 2 654 2 163 and adminis total operati 8 408 5 413 16 029 10 443 expenses operat 63 734 28 440 117 270 50 078 income other income 7 773 2 766 24 140 3 039 net income before 71 507 31 206 141 410 53 117 income tax income tax 11 819 4 784 23 400 7 920 expense net 59 688 26 422 118 010 45 197 income net income per share basic 2.47 1.08 4.87 1.85 diluted 2.46 1.08 4.85 1.84 weight average shares used in per share comput basic 24 190 24 366 24 238 24 404 diluted 24 285 24 532 24 338 24 571 nvidia corporation condensed consolidated balance sheets in millions unaudited assets july 26 january 25 current assets 2026 2026 cash and cash 22 443 10 605 equivalents marketable debt 34 143 39 065 securities marketable equity 42 783 12 886 securities accounts receivable 63 059 38 466 net inventories 31 575 21 403 prepaid expenses and 3 409 3 180 other current assets total current 197 412 125 605 assets property and 14 285 10 383 equipment net operating lease 5 390 2 867 assets goodwill 21 125 20 832 intangible assets 2 998 3 306 net deferred income tax 12 159 13 258 assets non marketable 51 157 22 251 securities other assets 15 746 8 301 total assets 320 272 206 803 liabilities and shareholders equity current liabilities accounts payable 15 059 9 812 accrued and other 26 960 21 352 current liabilities short term debt 1 000 999 total current 43 019 32 163 liabilities long term debt 32 366 7 469 long

term operating 4 985 2 572 lease liabilities other long term 10 918 7 306
 liabilities total liabilities 91 288 49 510 shareholders 228 984 157 293 equity
 total liabilities and shareholders 320 272 206 803 equity nvidia corporation
 condensed consolidated statements of cash flows in millions unaudited three
 months six months cash flows ended ended from operating july 26 july 27 july 26
 july 27 activities 2026 2025 2026 2025 net income 59 688 26 422 118 010 45 197
 adjustments to reconcile net income to net cash provided by operating activities
 stock based compensation 2 027 1 624 3 954 3 099 expense depreciation and 1 127
 668 2 124 1 280 amortization deferred 602 18 982 2 160 income taxes gains from
 equity 7 771 2 247 23 707 2 073 securities net other 315 100 222 196 changes in
 operating assets and liabilities net of acquisitions accounts 22 346 5 675 24
 590 4 743 receivable inventories 5 784 3 622 10 204 4 880 prepaid expenses and 5
 497 387 6 480 946 other assets accounts 1 915 1 314 4 125 2 255 payable accrued
 and other current 252 4 053 8 015 3 075 liabilities other long term 753 629 1
 970 979 liabilities net cash provided by 24 077 15 365 74 421 42 779 operating
 activities cash flows from investing activities proceeds from sales and
 maturities of 24 592 3 150 26 563 6 739 debt securities proceeds from sales of 7
 215 70 7 241 70 equity securities purchases of equity 15 822 346 42 404 1 245
 securities purchases of debt 21 777 7 812 21 777 14 108 securities purchases
 related to property and 2 677 1 894 4 434 3 122 equipment and intangible assets
 acquisitions net of cash 211 294 298 677 acquired other 15 15 net cash used in
 investing 8 695 7 126 35 124 12 343 activities cash flows from financing
 activities proceeds related to issuance of debt 24 896 24 896 net of costs
 proceeds related to employee stock plans 515 370 payments related to repurchases
 of 19 732 9 721 39 044 23 815 common stock dividends paid 6 047 244 6 290 488
 payments related to employee stock plan 2 402 1 848 4 531 3 380 taxes groq inc.
 2 944 2 944 principal payments on property and 59 21 92 73 equipment and
 intangible assets other 112 31 net cash used in financing activities 6 176 11
 834 27 459 27 386 change in cash and cash equivalents 9 206 3 595 11 838 3 050
 cash and cash equivalents at beginning 13 237 15 234 10 605 8 589 of period cash
 and cash equivalents at end of 22 443 11 639 22 443 11 639 period nvid corp reco
 of gaap to non fina gaap meas cost in of mill rev exce gaap per gro share pro
 data gaap una gro acqu mar and non other cost costs of a rev other non acqu gro
 and pro other non costs gro a mar other gaap total ope impa exp of non non ope
 adju exp to gaap oper ope inco 24 0 20 4 12 890 44 538 30 284 inc gains 72 1 61
 1 33 853 133 2 60 521 non from 75.0 74.9 72.4 75.0 66.6 ope equi 46 47 49 93 170
 inc secu 28 28 4 gaap net 24 0 20 3 12 841 44 417 30 110 oth other 72 1 61 2 33
 902 133 4 60 695 inc b 75.0 75.0 72.5 75.0 66.8 net total 8 408 7 621 5 413 16
 029 10 443 non pre 176 172 37 348 74 oth impa 15 15 inc of 8 232 7 449 5 361 15
 681 10 354 net non 63 7 53 5 28 440 117 2 50 078 gaap adju 222 247 101 469 263
 net inco 63 9 53 7 28 541 117 7 50 341 inc tax 7 773 16 3 2 766 24 140 3 039 non
 impa 7 7 15 2 24 23 7 2 07 net of 298 26 1 323 2 inc non three 300 457 520 six
 756 968 dil adju mont 59 6 58 3 26 422 mont 118 0 45 197 net tax ended 7 2 15 2
 14 ended 22 9 1 80 inc expe july 1 517 april 2 890 july 438 july 4 407 july 418
 per from 26 26 27 48 26 27 48 sha obbb 2026 53 9 2026 45 5 2025 24 763 2026 99
 502 2025 43 855 gaap 2.46 2.39 1.08 4.85 1.84 non 2.22 1.87 1.01 4.09 1.78 wei
 ave sha used in dil 24 2 24 3 24 532 24 338 24 571 net inc per sha com gaap net

cash provided by 24 077 50 344 15 365 74 421 42 779 operating activities purchases related to property and equipment 2 677 1 757 1 894 4 434 3 122 and intangible assets principal payments on property and equipment 59 33 21 92 73 and intangible assets free cash flow 21 341 48 554 13 450 69 895 39 584 includes h2o charges releases net which were billion and none for the first quarter and million and insignificant for the second quarter of fiscal years 2026 and 2027 respectively. tax expense included represents impact from obbba one big beautiful bill act . a acquisit and other costs are comprised of amortiza of intangib assets transact costs and certain compensa charges and are included in the following line items three six months months ended ended july 26 april 26 july 27 july 26 july 27 2026 2026 2025 2026 2025 cost of 46 47 49 93 170 revenue research and 170 167 29 337 57 developme sales general 6 5 8 11 17 and administr b comprised of net gains on equity derivati interest expense related to acquisit consider discount to be paid in the future share of net earning related to equity method investme and dividend income on equity securiti nvidia corporation reconciliation of gaap to non gaap outlook q3 fy2027 outlook in billions gaap gross margin 74.0 impact of acquisition related costs and other costs non gaap gross margin 74.0 gaap operating expenses 9.2 acquisition related costs and other costs 0.2 non gaap operating expenses 9.0',
 'nvidia_mentions': 6.251776072747941,
 'amd_mentions': 0.0,
 'intel_mentions': 0.0,
 'gpu_mentions': 1.7050298380221653,
 'capex_up_score': 0.0,
 'capex_down_score': 0.0,
 'competitor_mentions': 0.0,
 'capex_net': 0.0,
 'ai_sentence_ratio': 0.0}

```
[32]: import textwrap
print(textwrap.fill(nvda_tickers[0]['Text'][:2000], width=100))
```

nvidia announces financial results for second quarter fiscal 2027 revenue of billion up 106 from a year ago data center revenue of billion up 117 from a year ago santa clara calif. aug. 26 2026
 nvidia nasdaq nvda today reported revenue for the second quarter ended july 26 2026 of billion up 18 from the previous quarter and up 106 from a year ago. for the quarter gaap and non gaap gross margins were both 75.0 . gaap and non gaap earnings per diluted share were and respectively. ai has reached its inflection point. its doing useful work. its tokens are productive and profitable. now compute is revenue said jensen huang founder and ceo of nvidia. and demand is accelerating. this time last year one lab alone was driving the buildout today we have a golden age of new ai labs and

startups multiple frontier labs scaling in parallel a thriving open model ecosystem and physical ai coming online with strong momentum across the u.s. and around the world. the ai infrastructure buildout is at full steam. vera rubin now in full production was built to power exactly this moment. during the second quarter of fiscal 2027 nvidia returned approximately billion to shareholders in the form of shares repurchased and cash dividends. as of the end of the second quarter the company had approximately billion remaining under its share repurchase authorization. nvidia will pay its next quarterly cash dividend of per share on october 1 2026 to all shareholders of record on september 10 2026. q2 fiscal 2027 summary gaap in mill exce q2 q1 fy27 q2 fy26 q q y y earn fy27 per shar reve 18 106 gross 75.0 74.9 72.4 0.1 pts 2.6 marg pts oper 10 55 expe oper 19 124 inco net 2 126 inco dilu earn 3 128 per share non g in milli except q2 fy27 q1 fy27 q2 fy26 q q y y earni per share reven 18 106 gross 75.0 75.0 72.5 2.5 pts margin opera 11 54 expen opera 19 124 income net 18 118 income dilut earni 19 120 per share outlook nvidias outlook for the third quarter of fiscal 2027 is as follows revenue is expecte

```
[33]: # merge back to original dataframe
df = df.reset_index()

df = pd.merge_asof(
    df.sort_values('Date'),
    transcript_signals_df.reset_index().rename(columns={'date': 'Date'}),
    ↪sort_values('Date'),
    on='Date',
    by='Ticker',
    direction='backward'
)

with_domain_keywords = df.set_index(['Date', 'Ticker'])
with_domain_keywords[['gpu_mentions', 'capex_up_score', 'capex_down_score', ↵
    ↪'capex_net', 'competitor_mentions', 'ai_sentence_ratio']] = \
    with_domain_keywords[['gpu_mentions', 'capex_up_score', 'capex_down_score', ↵
    ↪'capex_net', 'competitor_mentions', 'ai_sentence_ratio']].fillna(0)
```

0.0.6 Final Cleaning

```
[34]: edgar_cols = ['gpu_mentions', 'capex_up_score', 'capex_down_score',  
                  'competitor_mentions', 'capex_net', 'ai_sentence_ratio']
```

```
[35]: with_domain_keywords[edgar_cols] = with_domain_keywords[edgar_cols].fillna(0)
```

```
[36]: with_domain_keywords.to_csv('../data/ai_sector_features_df.csv')
```

```
[37]: with_domain_keywords.head()
```

```
[37]:
```

		ret_1d	ret_3d	ret_5d	ret_10d	ret_20d	vol_5d	\
Date	Ticker							
2019-01-02	NVDA	0.020375	0.038500	0.071923	-0.051261	-0.166494	0.023554	
	TSM	-0.010566	-0.007069	0.034854	0.008282	-0.028465	0.020815	
	AVGO	-0.003028	0.010604	0.084303	0.008072	0.079067	0.025708	
	MRVL	0.019148	0.050955	0.145038	0.086345	0.028353	0.028901	
	INTC	0.003196	0.015530	0.080064	0.000000	-0.045224	0.024625	

		vol_20d	rsi_14	rel_soxx	rel_qqq	...	new_count	\
Date	Ticker					...		
2019-01-02	NVDA	0.034328	37.207888	-0.107088	-0.083557	...	0.0	
	TSM	0.017657	49.090091	0.030941	0.054471	...	0.0	
	AVGO	0.026895	58.335510	0.138473	0.162004	...	0.0	
	MRVL	0.030025	64.768966	0.087758	0.111289	...	0.0	
	INTC	0.025632	49.416392	0.014181	0.037712	...	0.0	

		nvidia_mentions	amd_mentions	intel_mentions	\
Date	Ticker				
2019-01-02	NVDA	0.626763	0.0	0.0	
	TSM	NaN	NaN	NaN	
	AVGO	0.000000	0.0	0.0	
	MRVL	NaN	NaN	NaN	
	INTC	0.000000	0.0	0.0	

		gpu_mentions	capex_up_score	capex_down_score	\
Date	Ticker				
2019-01-02	NVDA	0.0	0.0	0.0	
	TSM	0.0	0.0	0.0	
	AVGO	0.0	0.0	0.0	
	MRVL	0.0	0.0	0.0	
	INTC	0.0	0.0	0.0	

		competitor_mentions	capex_net	ai_sentence_ratio
Date	Ticker			
2019-01-02	NVDA	0.0	0.0	0.000000
	TSM	0.0	0.0	0.000000
	AVGO	0.0	0.0	0.000000

MRVL	0.0	0.0	0.000000
INTC	0.0	0.0	0.003636

[5 rows x 27 columns]

```
[38]: missing = set(all_tickers) - set(transcript_signals_df.index.
      get_level_values('Ticker').unique())
      print(missing)
      # {'ARM', 'TSM', 'ASML'}
```

```
{'ARM', 'TSM', 'ASML'}
```

```
[ ]:
```